Skills Summary

Equivalent Fractions and Comparing — Read Both Numbers

Use this reference while completing your worksheet or when studying.

Skill 1 — When the tops or the bottoms match

What each number does: the **bottom** tells how many equal pieces the whole was cut into, so a bigger bottom makes *smaller* pieces. The **top** tells how many pieces you have.

Bottoms match: the pieces are the same size, so the bigger top wins.

Tops match: you hold the same number of pieces, so the SMALLER bottom wins, because its pieces are bigger.

Neither matches: no shortcut. Go to Skill 3.

Example: Which is greater, $\frac{4}{9}$ or $\frac{4}{5}$?
Step 1. The tops match, so this is the same number of pieces in each fraction and only the size of a piece can decide it.
Step 2. A whole cut into 9 gives smaller pieces than a whole cut into 5, and four small pieces are less than four big ones. $\mathbf{\frac{4}{9} < \frac{4}{5}}$

Try it: Write $<$, $>$ or $=$ between $\frac{5}{6}$ and $\frac{5}{12}$.
check: $\mathbf{\frac{5}{6} > \frac{5}{12}}$

Skill 2 — Build an equal fraction

Rule: multiply the top AND the bottom by the same number, and the fraction keeps its value. Divide them both by the same number and it still does.

Changing only one of the two numbers does not make an equal fraction — it makes a different fraction altogether.

How to find the missing top: see what the bottom was multiplied by, then do that to the top.

Example: $\frac{2}{3}$ written with 12 on the bottom is $\frac{\square}{12}$. Find the top.
Step 1. Start from the bottom number, because it is the one that changed and it tells you what was done to the whole fraction.
Step 2. 3 was multiplied by 4 to reach 12, so the top is multiplied by 4 as well: 2×4. The missing top is **8**, so $\frac{2}{3} = \frac{8}{12}$.

Try it: $\frac{3}{4} = \frac{\square}{20}$. Find the missing top number.
check: **15**

Skill 3 — Rewrite both, then compare

When neither number matches: choose a bottom number that both bottoms divide into (multiplying the two bottoms together always works), rewrite each fraction with it using Skill 2, then compare the tops.

Once the bottoms are equal, the pieces really are the same size — and only then does counting tops mean anything.

Example: Which is greater, $\frac{2}{5}$ or $\frac{3}{8}$?

Step 1. Neither the tops nor the bottoms match, so nothing can be read off directly — rewrite both with one shared bottom number first.

Step 2. Both 5 and 8 divide into 40: $\frac{2}{5} = \frac{16}{40}$ and $\frac{3}{8} = \frac{15}{40}$, and sixteen pieces beat fifteen equal ones. $\boldsymbol{\frac{2}{5} > \frac{3}{8}}$

Try it: Write $<$, $>$ or $=$ between $\frac{5}{8}$ and $\frac{2}{3}$. (Try 24 on the bottom.)

check: $\boldsymbol{\frac{5}{8} < \frac{2}{3}}$

(!) Watch out: “bigger bottom number, bigger fraction” is backwards. More pieces always means smaller pieces — $\frac{1}{100}$ is tiny, not huge.